GIS ASSIGNMENT

Analysis on Household expenditure on Services and Durable goods - JULY 2014 - JUNE 2015, 72 round.

State: Sikkim

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Second Year

PLAGIARISM REPORT

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INTRODUCTION

Household expenditure plays a pivotal role in understanding the socio-economic dynamics of a society. This is because, despite being under the economic domain, expenditure also gets governed by a lot of social elements which puts it under the two major sheds for understanding society. Sikkim, being the smallest state in India by population and second smallest state by size, has a unique geographical and cultural characteristic due to its location. It is also one of the busiest states in terms of tourism. This makes it important to understand the socio-economic domains at an overall level essential for developing targeted policies and interventions.

This study focuses on two important aspects of household expenditure using data from the National Sample Survey (NSS) 72nd round (2014-15): domestic services and business services. Analyzing expenditures at domestic levels and finding if the gap exists between urban and rural sectors for business level expenditure will aim to uncover meaningful trends for economic planning in Sikkim and understanding how it governs?

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INTRODUCTION

Household expenditure plays a pivotal role in understanding the socio-economic dynamics of a society. This is because, despite being under the economic domain, expenditure also gets governed by a lot of social elements which puts it under the two major sheds for understanding society. Sikkim, being the smallest state in India by population and second smallest state by size, has a unique geographical and cultural characteristic due to its location. It is also one of the busiest states in terms of tourism. This makes it important to understand the socio-economic domains at an overall level essential for developing targeted policies and interventions.

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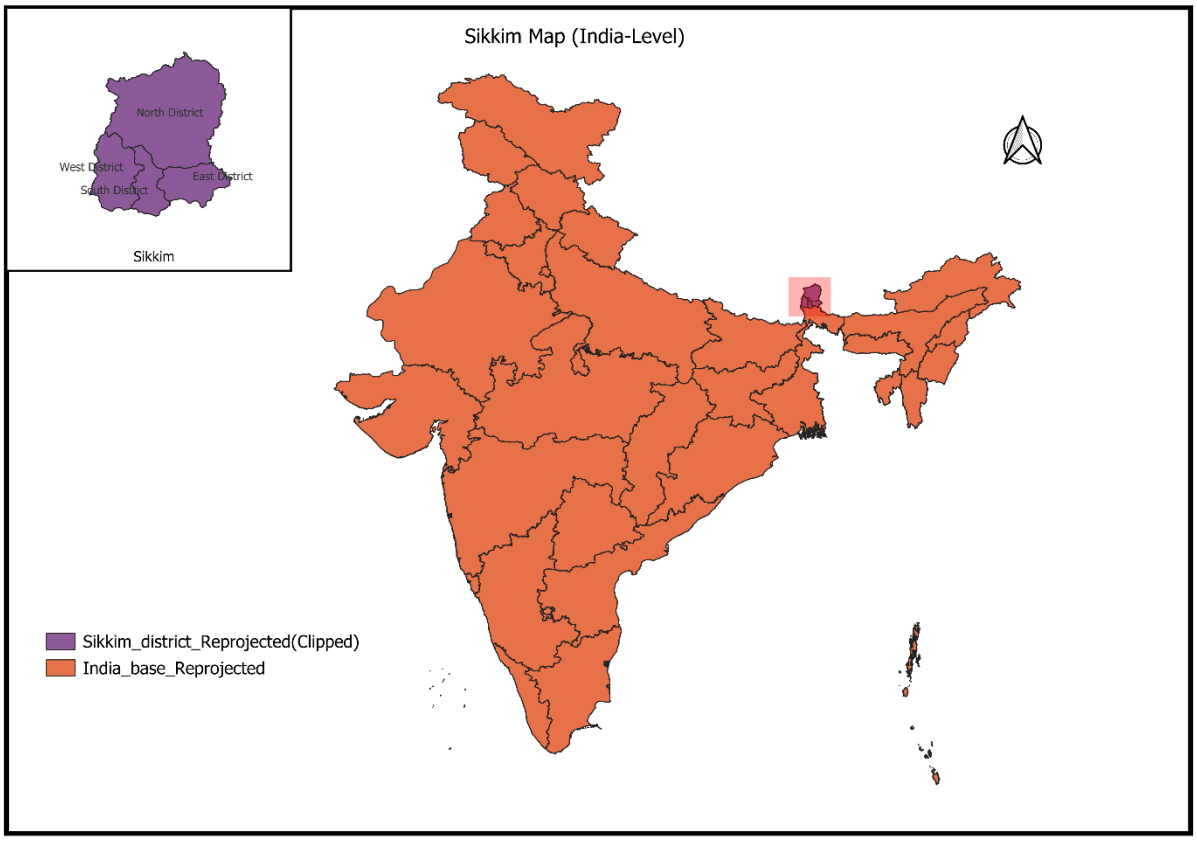


METHODOLOGY

Study Area

The area under our research is Sikkim. Sikkim is one of the smallest states in India with its unique geographical features and diverse socio-economic characteristics. It is sub-divided into four districts, along with their own distinct urban and rural settlements and is what makes it ideal for analyzing household expenditure patterns.

Dataset:

The data set chosen is from the National Sample Survey (NSS) 72nd round (2014-15) which covers household expenditure on various services. Some variables under central analysis include Household

size, Expenditure on domestic services (530-533), Expenditure on business services (630-632) and Sector. The primary unit of analysis is the household and visualizations are created on the basis of state and district level shape files.

Research methods:

The NSS data was filtered to include only relevant data on Sikkim. After that, data corresponding to mentioned variables were extracted. Statistical summaries such as mean, median, correlation, standard deviation was found to get an overview of the data. Maps were created to get initial visualizations as well as represent spatial patterns of household expenditure across Sikkim. MS Excel was used as a primary tool for analysis. Again, district-level shape files were used to depict trends.

Research Question:

- How does the total household expenditure on domestic services correlate with household size in Sikkim?
- What is the relationship between household expenditure on business services and the urban-rural divide in Sikkim?

ANALYSIS AND DISCUSSION

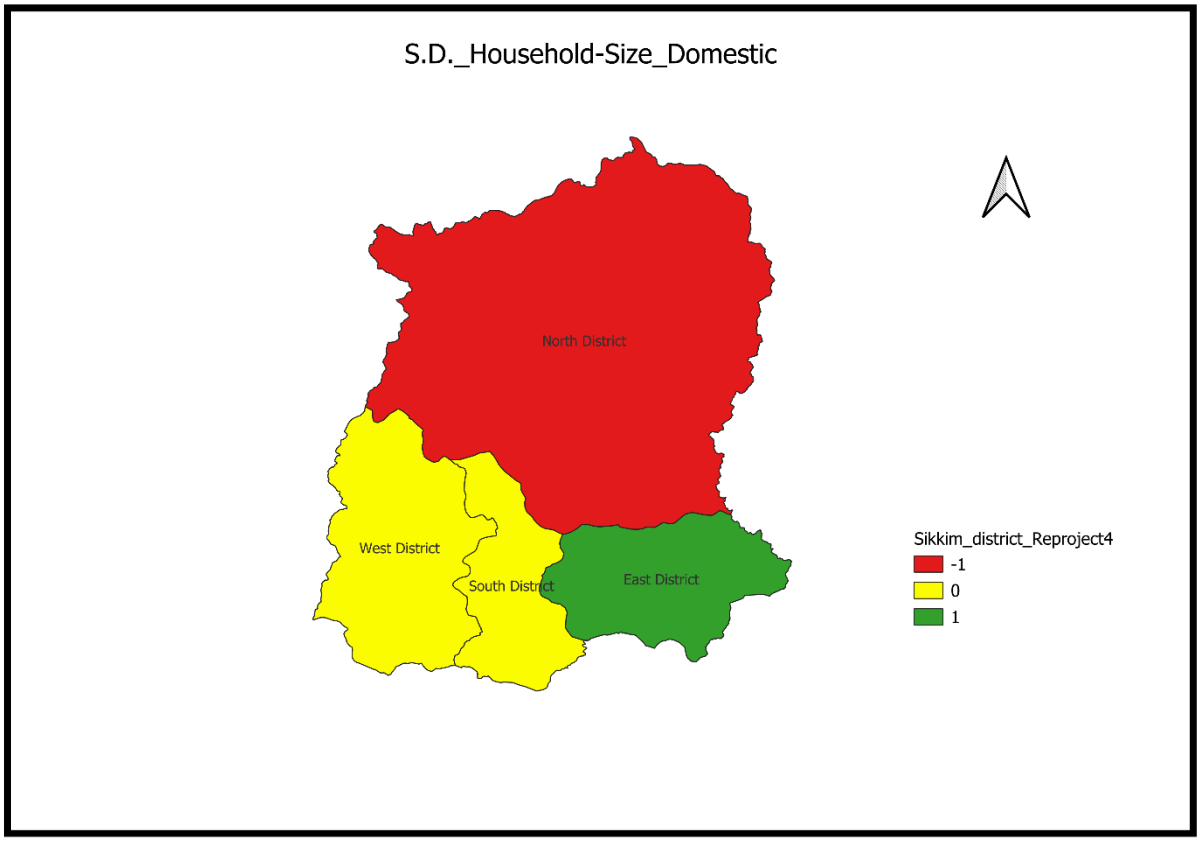
Research question 1

At first this study is focused to find out whether household expenditure on domestic services show any correlation with the household size. This becomes important because how household takes decisions at domestic levels will give a broader picture of their mentality that what is their primary focus area for managing their own needs. What is priority for the population and is there a coordination among household members for managing services while sharing resources in a household or it gets managed through expenditure. And focus is on availing quality domestic services. A district-level was analysis was done to locate the areas in Sikkim which need focus for development in their quality of life and economic growth.

Analysis 1.1

While studying households involved in domestic services, it was found that there was a significant deviation noticed among districts in context of household size. With the average household size of 1 North district was showing negative deviation which was largest in size but a positive deviation with an average household size of 3 was observed in East Sikkim. While West and South districts shown no deviation signifying average household size of 2. The average household size in Sikkim was 2 *to be accurate* (1.86). This might be due predominance of rough terrain in Northern district and also presence of protected areas like Khangchendzongha National Park. Presence of higher number of indigenous communities might also be a reason for this less average population. While, better connectivity and absence of these, might be a reason for a higher household size in spite being relatively lower in size.

When analyzed the deviation in context of expenditure among districts, least deviation was found in Northern and Western districts which might be due to rough terrain areas and areas that are more focused on agricultural. While Eastern districts came up as the area which shows highest deviation in expenditure among domestic services. This might be due to the presence of major urban centers in the area.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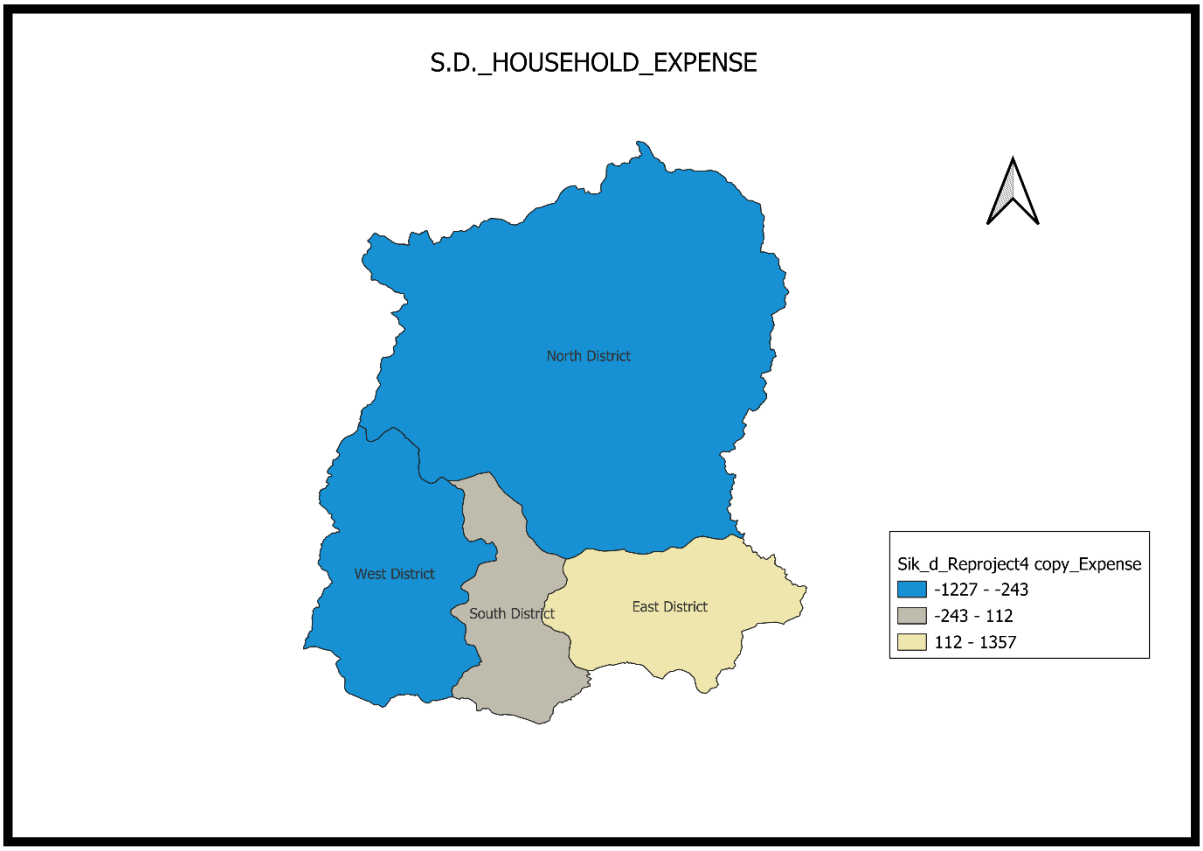
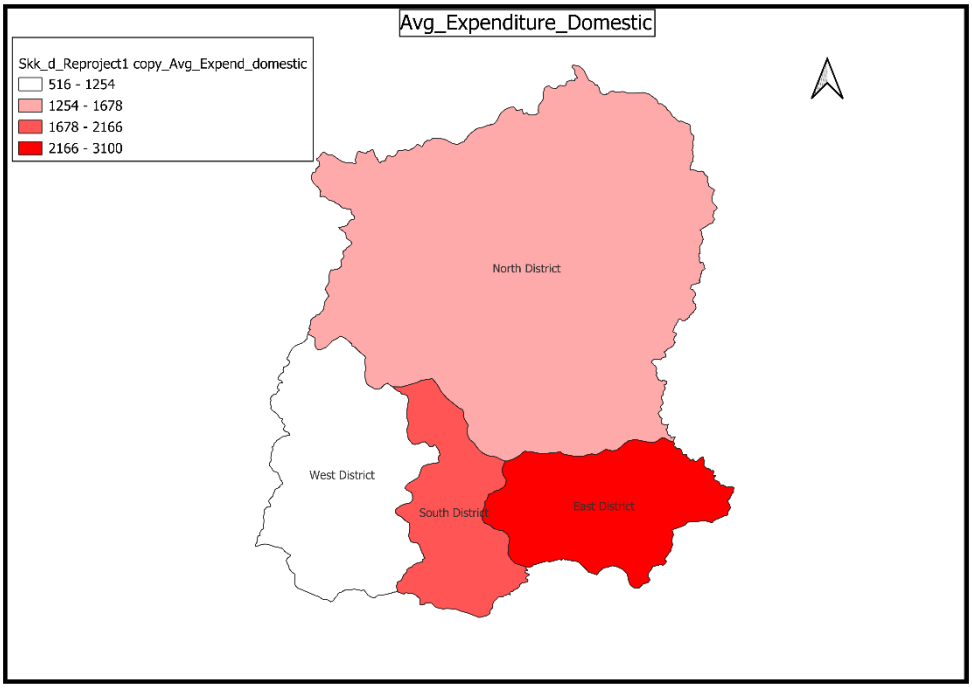
North District
West District
South District
East District
Sikkim_district_Reproject4
-1
0
1

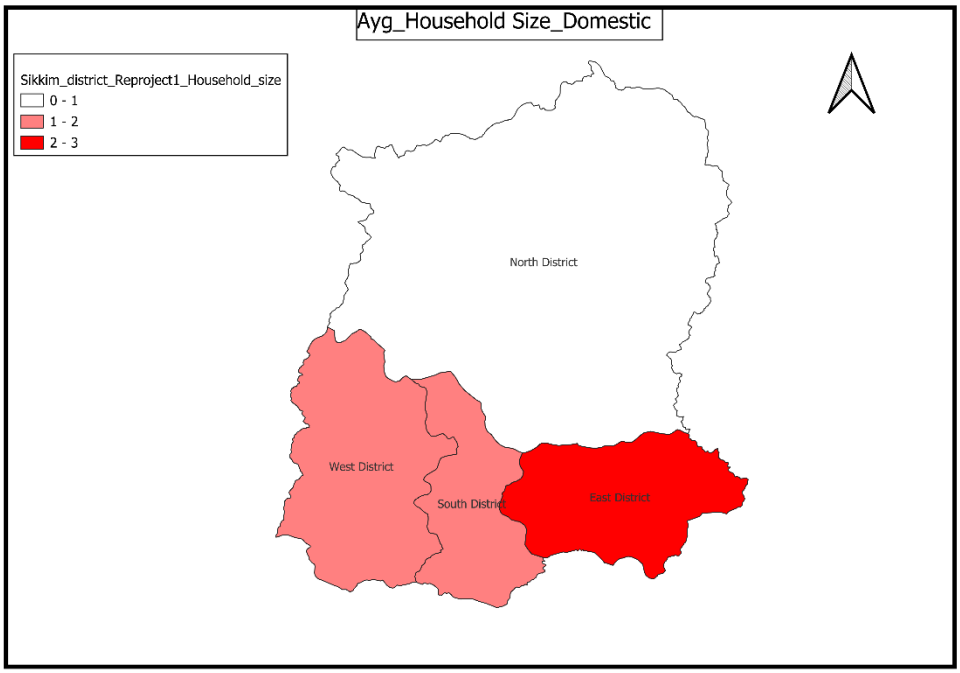

S.D._HOUSEHOLD_EXPENSE
North District
West District
South District
East District
Sik_d_Reproject4 copy_Expense
-1227 - -243
-243 - 112
112 - 1357

Analysis 1.2

Through this an area of interest which was visible for me is there a correlation between the household size and related expenditure on domestic services which was essential to understand the working style of the people. If there is a positive correlation it suggests that people are more involved in their own social chores and more inclined towards inclination of availing domestic services for their needs. While this is not the case in a negative correlation where it would signify people share the responsibilities among household members hence, decreasing the overall expenditure in domestic services.

Below are the graphs made on the basis of average household expenditure and average household size among districts and the data within it are classified on the basis of equal counts or quantile sizes.



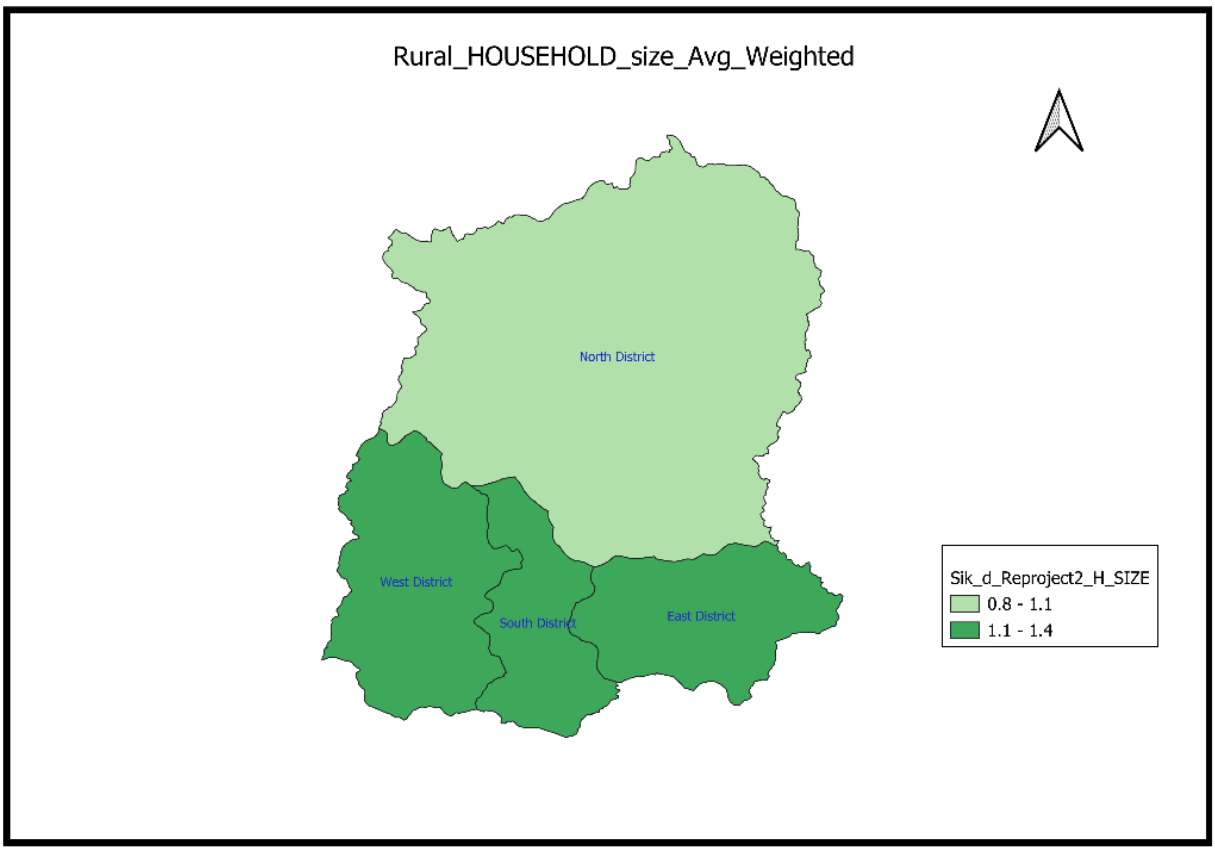


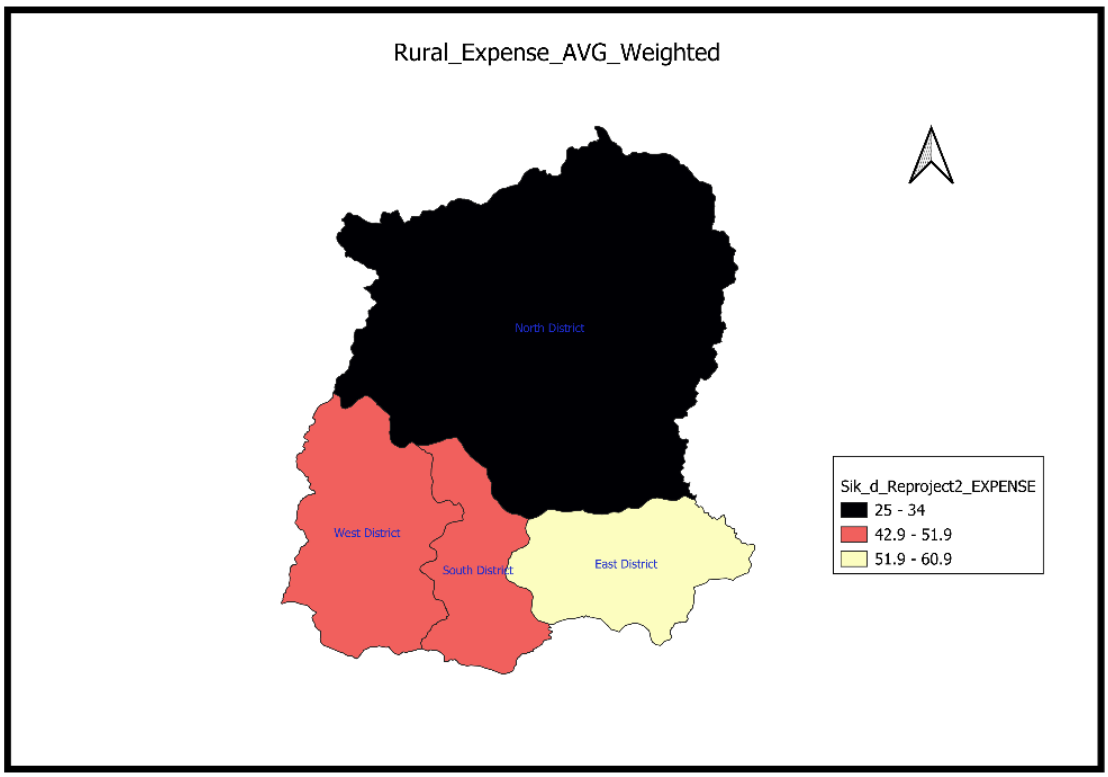
Through an analysis on Excel, when household sizes and expenditure were run under correlation analysis. It showed a correlation of 0.77 which signifies a strong correlation showing how with an increase in household size the expenditure on domestic services also increases. Through, close analysis of the maps we can also find how the East district shows highest household expenditure and household size too. While North and west are at lower levels in both cases with South at second highest average in both cases. Interestingly, it sounds quite obvious that with the increase in household size we see an increase in expenditure but it also underlies how work distribution works and what role it plays at individual levels or in other words, do people have space and time to devote in their domestic chores? It also signifies other social factors in it like availability of such services and prevalent traditional and cultural norms.

A visibly strong correlation is present among the districts for household expenditures and sizes. But this sparks a question to understand if there was a less involvement of individuals for domestic services then what is its scenario at urban and rural levels. And is there any rural urban divide governs these. What is the relationship between household expenditure on business services and the urban-rural divide in Sikkim?

Research question 2

To find it the analysis was divided at two levels. First focusing on business areas and then, understanding it at urban and rural levels in a district wise manner.



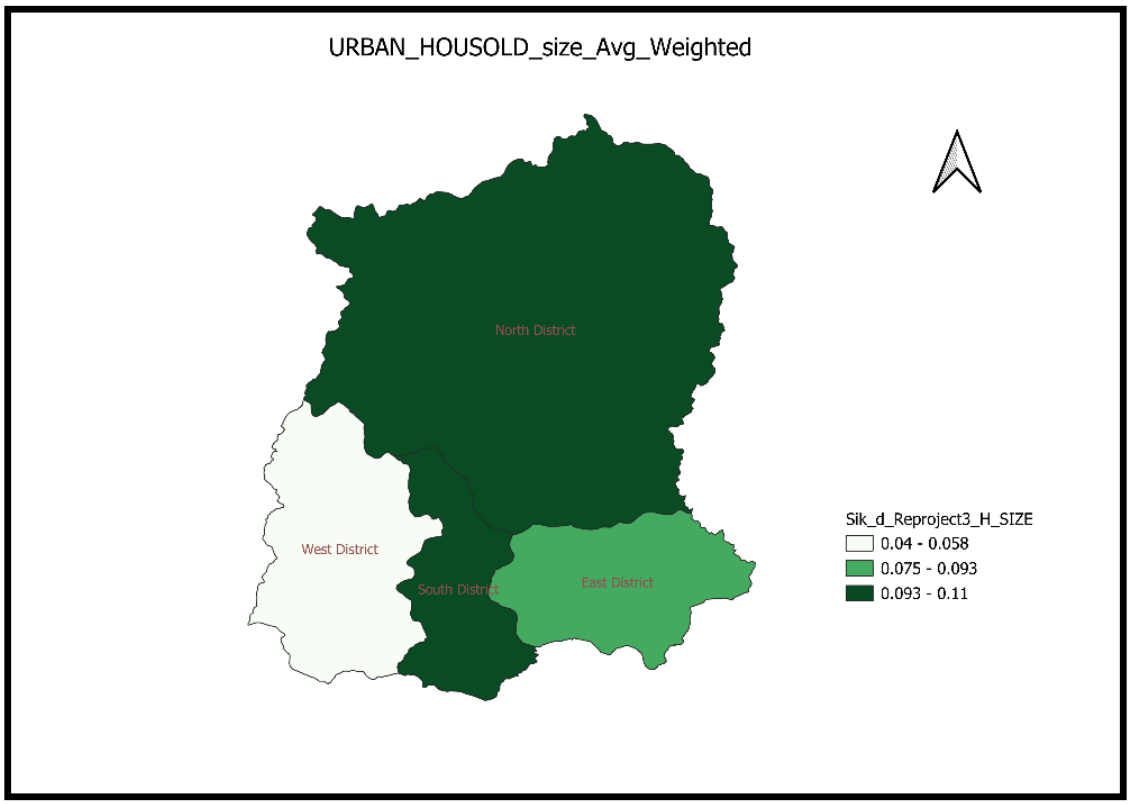


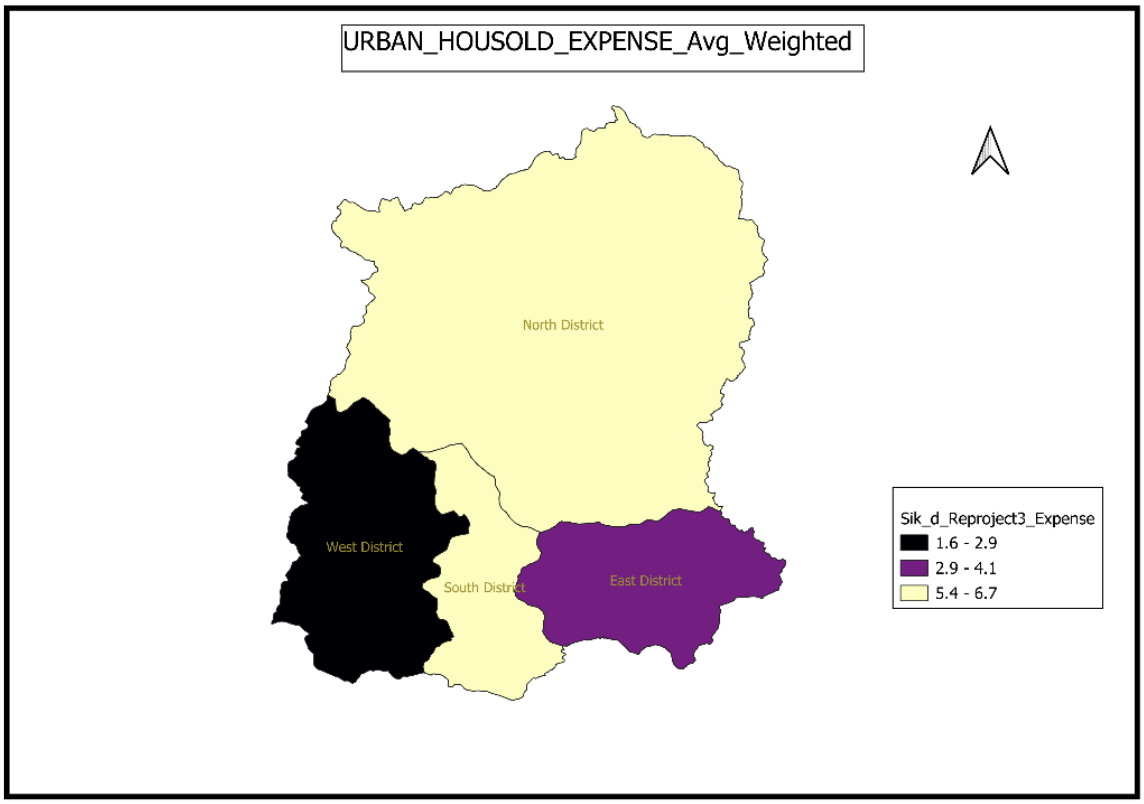
To analyze this, we took a close look to weighted average which was calculated for each district on the basis of their separate rural and urban setups. For instance, if we have to calculate the weighted expenditure of rural area in East district has with an average expenditure of Rs.66. This would be calculated by:

Weight= (No. of Rural Households in East District / No. of Households in East District)

Weighted Expenditure= Weight * Average expenditure in rural area of East District

It was found that the average weighted household size was noted to be least in North districts whereas it was comparatively average in other districts. Broadly speaking, the weighted average household size did not show much variation. It varied with a range of 0.39. The rural weighted expense was least in North district and highest in East districts while it was comparatively average in the two other districts. In context of rural setup, The North District displayed rugged terrain and thus, a justified with less average weighted rural household size and business expense was noticed. While a very slight change in household size was noticed in other rural areas of Sikkim, its average weighted expenditure varied. The areas with the scope of better opportunities like East district showed better investment in business expenditures.


URBAN_HOUSOLD_size_Avg_Weighted
North District
West District
South District
East District
Sik_d_Reproject3_H_SIZE
0.04 - 0.058
0.075 - 0.093
0.093 - 0.11


URBAN_HOUSOLD_EXPENSE_Avg_Weighted
North District
West District
South District
East District
Sik_d_Reproject3_Expense
1.6 - 2.9
2.9 - 4.1
5.4 - 6.7

In the urban setup, the scenario is a little different. In urban area the districts with highest average weighted household number has highest weighted average expenditure which is visible in case of North and South districts while similar case is seen with the East district with highest average household number and weighted expense.

So, what we infer is that urban people are more inclined in business services thus, a positive correlation between the household size and expenditure is notices. While in rural areas this was not the case. People were not that much invested in the business areas in rural areas, the household number did not affect the expenditures on business, possibly due to dominance of urban points in businesses making it hard for rural and points to grow. Also, lesser infrastructures and opportunities in rural areas might be a major factor of influence.

CONCLUSION

Conclusively, a strong correlation between household size and domestic service expenditure indicates larger the household more the expenditure on domestic services, of which availing and a need for a better domestic service might come up as a reason. A prominent rural-urban divide was visible in the field of business services. In urban areas, household size is positively correlated with business services. Economic flexibility of urban households, driven by smaller households' sizes and access to services is reflected by this correlation. Whereas, rural areas don't show a such strong inclination, this shows that they might be more inclined towards domestic services while some might get affected due to geographical, infrastructural or economic constraints.

Policies centered for improving services domestic services in rural areas especially in North and West Sikkim might aid in improving service availability at domestic services. Targeted interventions are needed to bridge rural-urban divide especially in business areas. Boosting rural business opportunities, among districts with higher households and using East Sikkim's strengths to model strategies might aid in bridging divide. Bringing the two sectors, at one page might address disparities among services and foster balanced development.

LIMITATIONS

The sample data used for analysis had a higher inclination towards rural sector with less data availability on urban sectors. Weighted averages were used to deal with less data availability. Also, the inferences generated through this is based on primary analysis and on a limited sample. Thus, the findings are totally based on this sample and is open for further interpretations and optimistic towards challenges.

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